

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

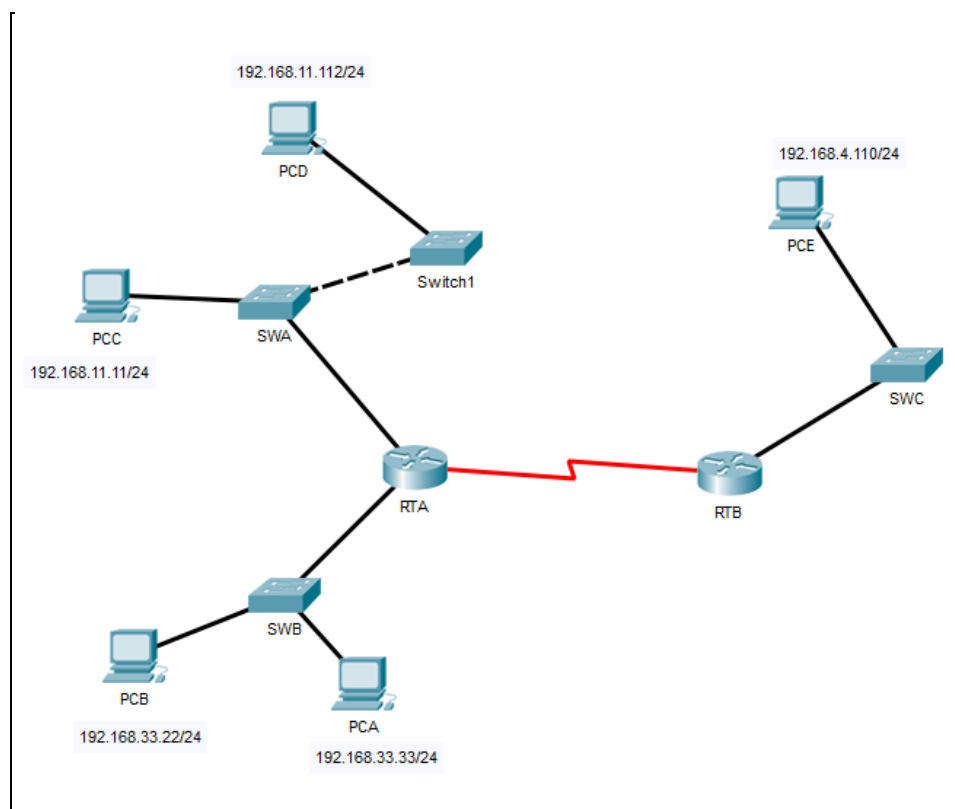


Figure 1

Part 1: Review the topology

Step 1: Open the *Lab_4_ARP.pkt*, and Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1      -          0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

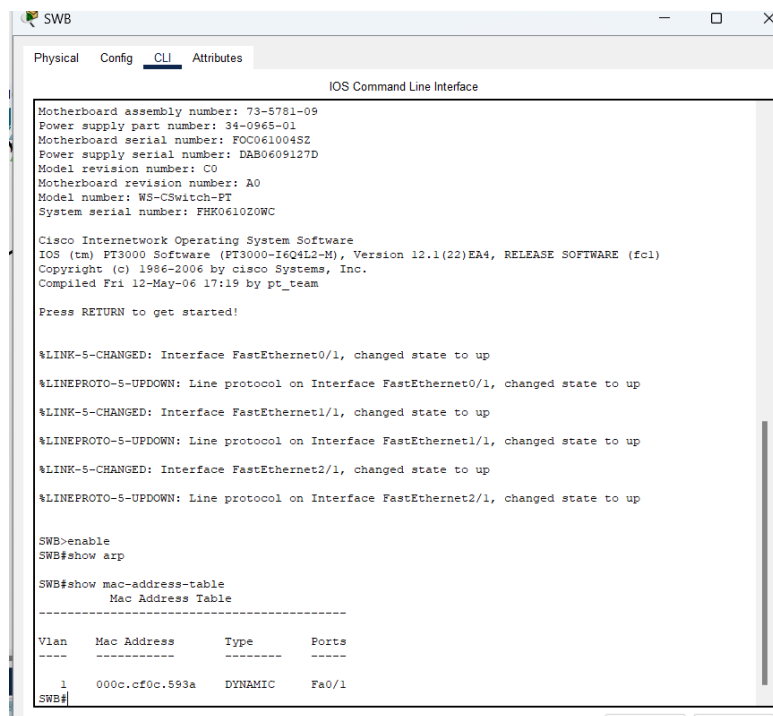
```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0002.4a00.0e91    DYNAMIC   Fa0/1
1     000c.8546.7d85    DYNAMIC   Fa1/1
SWA#
```

SWB



```
SWB
Physical Config CLI Attributes
IOS Command Line Interface

Motherboard assembly number: 73-5781-09
Power supply part number: 34-0965-01
Motherboard serial number: FOC061004SZ
Power supply serial number: DAB0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: WS-CSwitch-PT
System serial number: FHK061020WC

Cisco Internetwork Operating System Software
IOS (tm) PT3000 Software (PT3000-I6Q4L2-M), Version 12.1(22)EA4, RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2006 by cisco Systems, Inc.
Compiled Fri 12-May-06 17:19 by pt_team

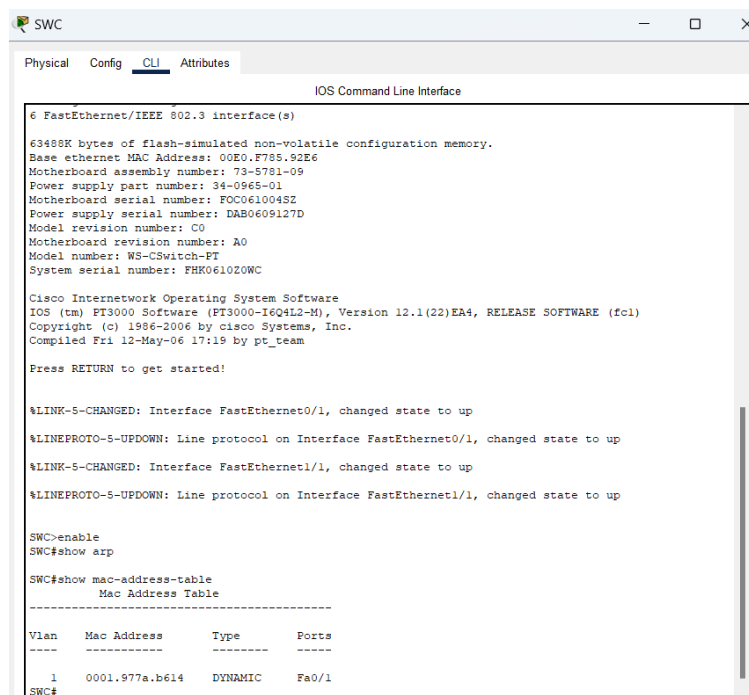
Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet2/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up

SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       000c.cf0c.593a   DYNAMIC   Fa0/1
SWB#
```

SWC



```
SWC
Physical Config CLI Attributes
IOS Command Line Interface

6 FastEthernet/IEEE 802.3 interface(s)
63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address: 00E0.F785.92E6
Motherboard assembly number: 73-5781-09
Power supply part number: 34-0965-01
Motherboard serial number: FOC061004SZ
Power supply serial number: DAB0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: WS-CSwitch-PT
System serial number: FHK061020WC

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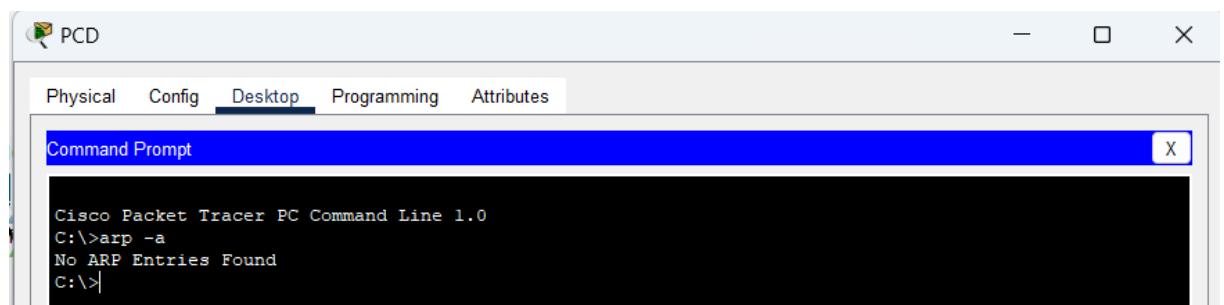
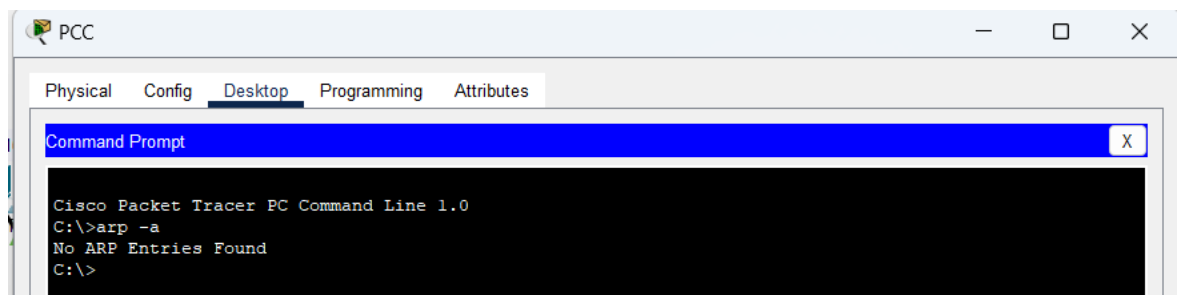
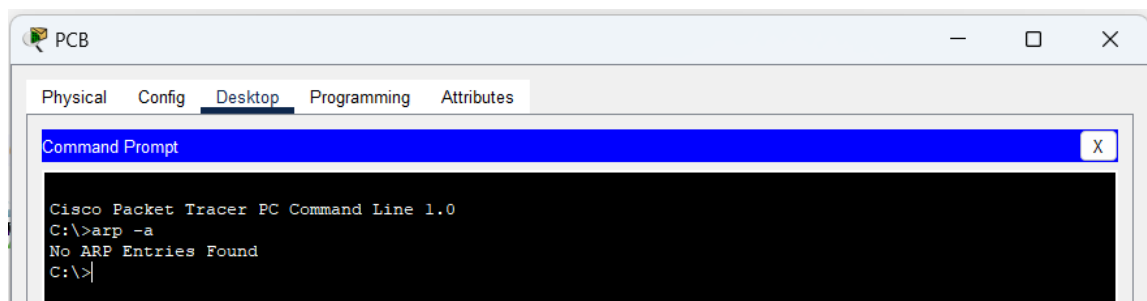
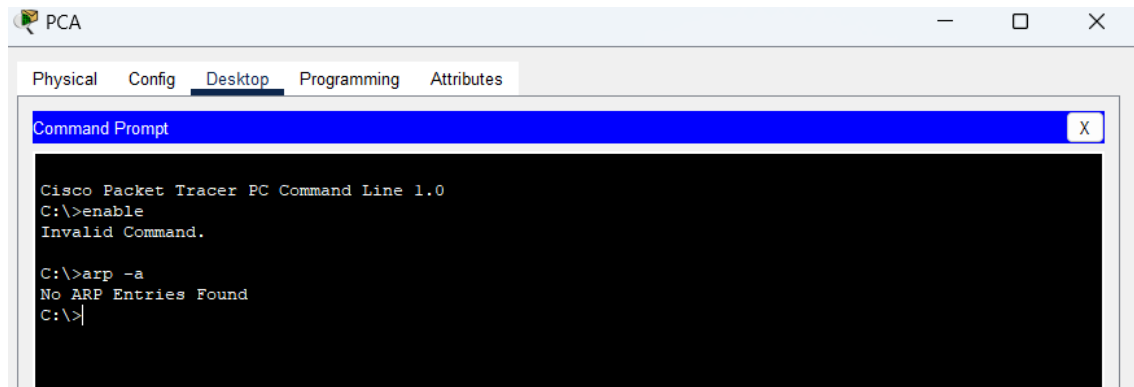
Press RETURN to get started!

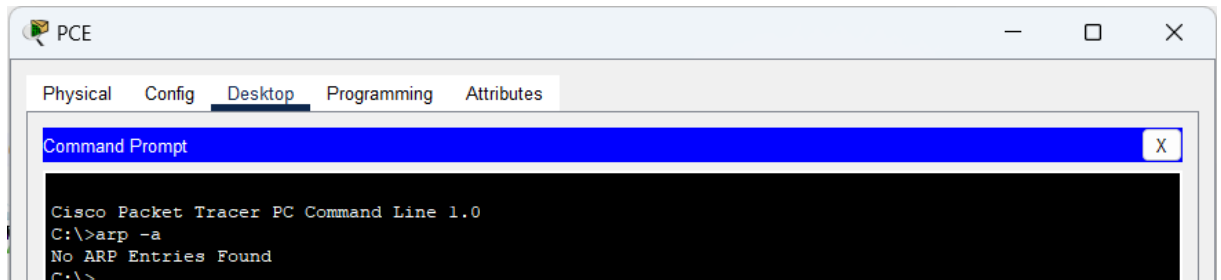
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up

SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.977a.b614   DYNAMIC   Fa0/1
SWC#
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.





- e. What are your thoughts on the results?

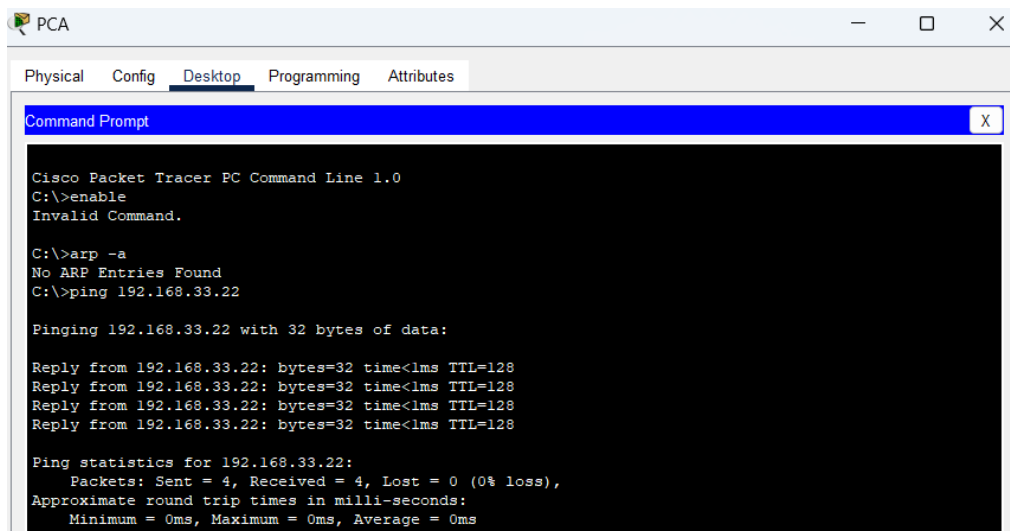
No ARP Entries Found on any PCs because all PC did not receive any ARP request. The request will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.



- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.33         0002.1755.9a06        dynamic
C:\>
```

- e. In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

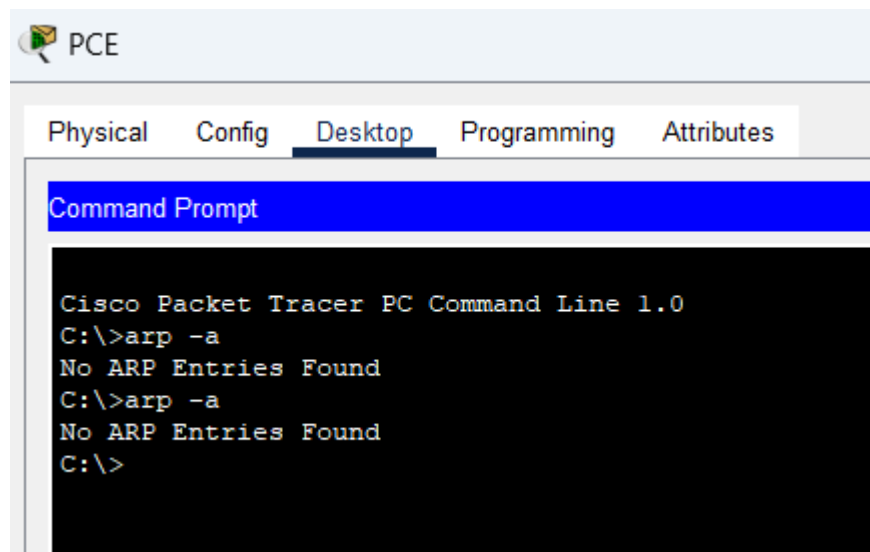
PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

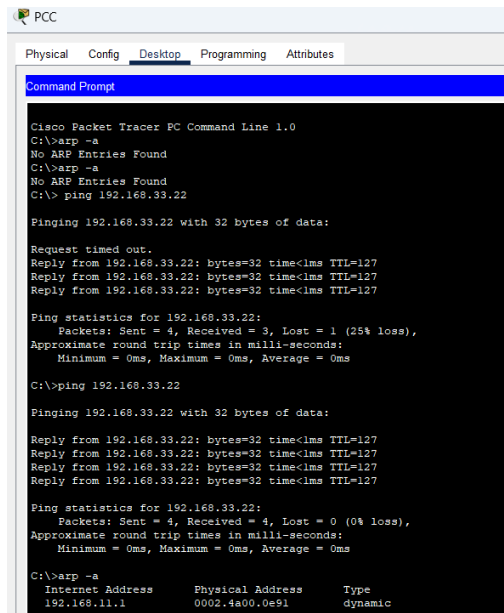
```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE



Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.



- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

    Internet Address      Physical Address        Type
    192.168.11.1          0002.4a00.0e91         dynamic
    192.168.11.112        0001.6462.0278         dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 7ms

C:\>arp -a

    Internet Address      Physical Address        Type
    192.168.11.1          0002.4a00.0e91         dynamic
    192.168.11.112        0001.6462.0278         dynamic
```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are in the ARP table only after PCC ping to PCB, PCD and PCE. No ARP entry before ping process. The ARP table displays entries once PCB, PCD and PCE retrieve and receive the ARP request from PCC.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.11.11 21 00D0.D39A.C0D9 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
Internet 192.168.33.22 21 0060.47EA.A746 ARPA FastEthernet0/0
RTA#
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 17 0060.702D.7C08 ARPA FastEthernet0/0
RTB#
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.4a00.0e91   DYNAMIC Fa0/1
1       000c.8546.7d85   DYNAMIC Fa1/1
SWA#
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
SWC#
```

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0005.5e95.187b   DYNAMIC Fa0/1
```

- d. Do switches use arp table? (Y/N) Y
- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation.*

It is not part of NetComm syllabi, it is just for knowledge..

Switches use ARP table to store ARP address and MAC address for network devices. Only layer 3 switches utilize ARP tables because ARP is layer 3 protocol that resolves IP addresses to MAC addresses for routing between subnets. In contrast, basic Layer 2 switches use only MAC address tables at the Data Link Layer to forward frames within the same network segment, so do not utilize ARP tables.

- f. What information does the command `show mac-address-table` gives?

The command `show mac-address-table` provides information about the MAC address table maintained by a switch. This table is used to associate MAC addresses with specific VLANs, port numbers, and types of learning (e.g., dynamic or static). The details typically include

1. **VLAN:** The VLAN ID associated with the MAC address.
2. **MAC Address:** The unique hardware address of the device connected to the switch.
3. **Type:** Indicates how the MAC address was learned:
 - a. **Dynamic:** Learned automatically by the switch through traffic.
 - b. **Static:** Manually configured by the network administrator.
4. **Ports:** The physical port(s) on the switch where the MAC address is connected.

Part 3: Attach wireless lab results.

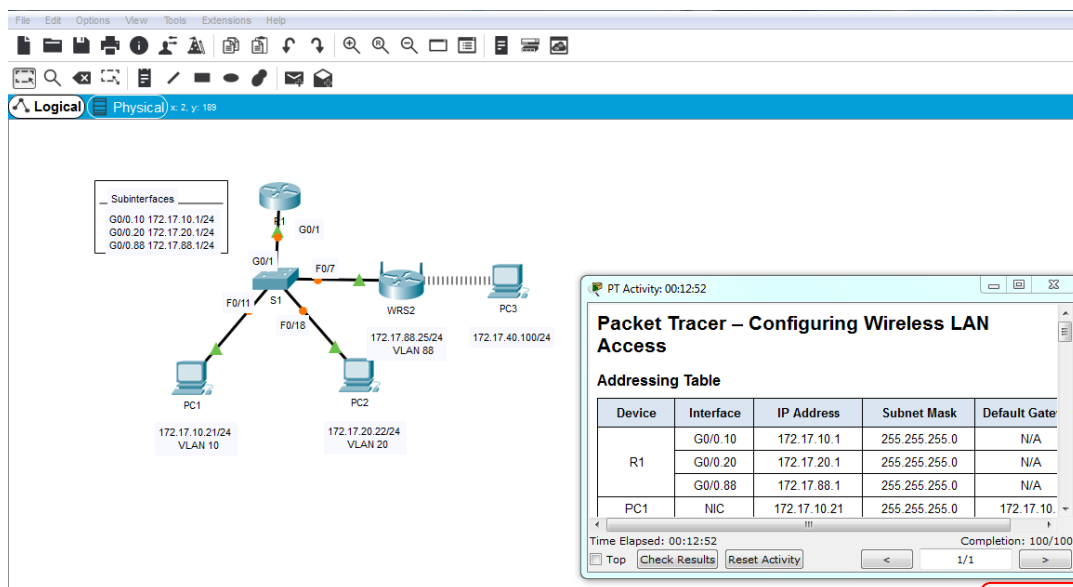
In this part, you will use **Lab_4_Wireless.pka** file.

Step1: Change the filename of the pka file.

- a. Change the Lab 4 filename to include your name. *Example: *Lab4AliAhmad.pkt*



- b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.



- c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Packet Tracer – Configuring Wireless LAN Access

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
WRS2	NIC	172.17.88.25	255.255.255.0	172.17.88.1

Objectives

Part 1: Configure a Wireless Router

Part 2: Configure a Wireless Client

Part 3: Verify Connectivity

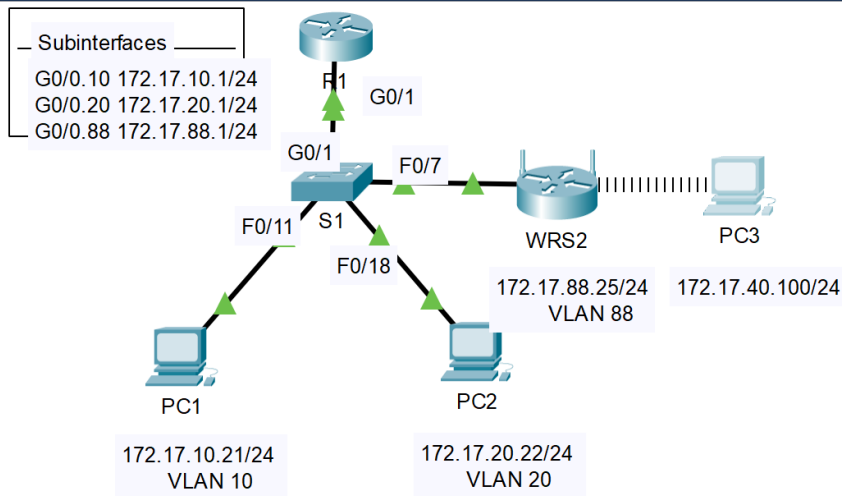
Scenario

In this activity, you will configure a wireless router, allowing for remote access from PCs as well as wireless connectivity with WPA2 security. You will manually configure PC wireless connectivity by entering the wireless router SSID and password.

Part 1: Configure a Wireless Router

Step 1: Connect the Internet interface of WRS2 to S1.

Connect the **WRS2** Internet interface to the **S1** F0/7 interface.



Step 2: Configure the Internet connection type.

- Click **WRS2 > GUI** tab.
- Set the **Internet Connection type** to **Static IP**.
- Configure the IP addressing according to the Addressing Table.

Wireless-N Broadband Router

Setup Wireless Security Access Restrictions Applications & Gaming Administration

Internet Setup

Internet Connection type: Static IP

Internet IP Address: 172 . 17 . 88 . 25

Subnet Mask: 255 . 255 . 255 . 0

Default Gateway: 172 . 17 . 88 . 1

DNS 1: 0 . 0 . 0 . 0

DNS 2 (Optional): 0 . 0 . 0 . 0

DNS 3 (Optional): 0 . 0 . 0 . 0

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Step 3: Configure the network setup.

- Scroll down to **Network Setup**. For the **Router IP** option, set the IP address to **172.17.40.1** and the subnet mask to **255.255.255.0**.
- Enable the DHCP server.
- Scroll to the bottom of the page and click **Save Settings**.

Network Setup

Router IP
IP Address: 172 . 17 . 40 . 1
Subnet Mask: 255.255.255.0

DHCP Server Settings
DHCP Server: ☒ Enabled ☐ Disabled
Start IP Address: 192.168.1.100
Maximum number of Users: 50
IP Address Range: 192.168.1.100 - 149
Client Lease Time: 0 minutes (0 means one day)
Static DNS 1: 0 . 0 . 0 . 0
Static DNS 2: 0 . 0 . 0 . 0
Static DNS 3: 0 . 0 . 0 . 0
WINS: 0 . 0 . 0 . 0

Step 4: Configure wireless access and security.

- At the top of the window, click **Wireless**. Set the **Network Mode** to **Wireless-N Only** and change the SSID to **WRS_LAN**.
- Disable **SSID Broadcast** and click **Save Settings**.
- Click the **Wireless Security** option.
- Change the **Security Mode** from **Disabled** to **WPA2 Personal**.
- Configure **cisco123** as the passphrase.
- Scroll to the bottom of the page and click **Save Settings**.

Wireless **Setup** **Wireless** **Security** **Access Restrictions** **Applications & Gaming** **Admin**

Basic Wireless Settings Wireless Security Guest Network Wireless MAC Filter

Basic Wireless Settings

Network Mode: Wireless-N Only
Network Name (SSID): WRS_LAN
Radio Band: Auto
Wide Channel: Auto
Standard Channel: 1 - 2.412GHz
SSID Broadcast: ☐ Enabled ☒ Disabled

Wireless		Setup	Wireless	Security	Access Restrictions	Applications & Gaming	Administration	Status
		Basic Wireless Settings		Wireless Security	Guest Network	Wireless MAC Filter	Advanced Wireless Settings	
Wireless-N Broadband Router WRT300N								
Wireless Security								
Security Mode:		WPA2 Personal						
Encryption:		AES						
Passphrase:		cisco123						
Key Renewal:		3600		seconds				
		Help...						

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.

Because SSID broadcast is disabled, you must manually configure **PC3** with the correct SSID and passphrase to establish a connection with the router.

- Click **PC3 > Desktop > PC Wireless**.
- Click the **Profiles** tab.
- Click **New**.
- Name the new profile **Wireless Access**.
- On the next screen, click **Advanced Setup**. Then manually enter the SSID of **WRS_LAN** on **Wireless Network Name**. Click **Next**.
- Choose **Obtain network settings automatically (DHCP)** as the network settings, and then click **Next**.
- On **Wireless Security**, choose **WPA2-Personal** as the method of encryption and click **Next**.
- Enter the passphrase **cisco123** and click **Next**.
- Click **Save** and then click **Connect to Network**.

Creating a Profile

Wireless Mode

Please choose the Wireless Mode that best suits your needs.

☒ **Infrastructure Mode**

Select Infrastructure Mode if you want to connect to a wireless router or access point.

☐ **Ad-Hoc Mode**

Select Ad-Hoc Mode if you want to connect to another wireless device directly without using a wireless router or access point.

Please enter the wireless network name (SSID) for your wireless network.

The wireless network name is shared by all devices in a wireless network and is case-sensitive.

Wireless Network Name

[Back](#) | [Next](#)

Wireless-N Notebook Adapter

Wireless Network Monitor v1.0

Model No. **WPC300N**

Creating a Profile

Wireless Security - WPA2 Personal

Pre-shared Key

Please enter a Pre-shared Key that is 8 to 63 characters in length.

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Wireless-N Notebook Adapter

Wireless Network Monitor v1.1

Model No. **WPC300N**



Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The **Signal Strength** and **Link Quality** indicators should show that you have a strong signal.
- Click **More Information** to see details of the connection including IP addressing information.
- Close the **PC Wireless** configuration window.

Part 3: Verify Connectivity

All the PCs should have connectivity with one another.

